SMART MOVE – Intelligent Daily Mobility Planner

Introduction

Urban mobility has become increasingly complex due to traffic congestion, unpredictable conditions, and multiple transport choices. SMART C is a responsive web application designed to help passengers and drivers make intelligent travel decisions using real-time simulation and multi-factor analysis. To address these challenges, we propose SMART C – Intelligent Daily Mobility Planner, a responsive web-based platform that assists both passengers and drivers in making informed travel decisions using multi-factor analysis.

Problem Statement

THEME : MOBILITY

The objective is to build a responsive web application that helps users make better day-to-day travel decisions by comparing different travel options based on: Time, Cost, Distance, Safety, Convenience, Real-world conditions (traffic, weather, peak hours)

Key Issues Identified:

- Lack of integrated comparison across multiple transport modes
- Limited consideration of real-world dynamic factors
- No support system for drivers to optimize routes and earnings
- We have to use multiple applications while travelling

3. Proposed Solution

SMART C is a dual-mode intelligent mobility platform consisting of:

Passenger Mode

Allows users to: Compare travel options, Analyze routes based on multiple factors, Receive smart recommendations, Access safety and feedback features.

Driver Mode

Provides tools to:Analyze ride profitability, Identify high-demand zones, Plan daily operations, Track earnings and performance.

Core Functionality

The system simulates real-world travel scenarios and provides optimized recommendations rather than just displaying raw data.

4. System Features

1 Passenger Features

Travel Input System

Source and destination entry

Transport Mode Selection

Public Transport (Bus, Metro), Rent a Vehicle, Hire a Driver, Own Vehicle.

Route Comparison

Displays:Multiple route options, Estimated time, Cost, Distance,Traffic conditions.

Smart Recommendation Engine

Highlights best travel option

Provides reasoning based on:Time, Cost, Traffic, Convenience.

Real-Time Simulation Panel

Simulates:

Traffic (Low / Medium / High), Weather (Sunny / Rainy), Peak Hour conditions.

Sustainability Panel

CO_2 emission estimation and Eco-friendly travel suggestions.

Safety Features

Emergency SOS button and Emergency contact display.

Feedback & Reporting System

Report issues (driver, vehicle, route), Submit feedback.

Commute History

Stores recent searches using Local Storage

Driver Features

Customer Reporting System

Report issues such as: Late payment, Cancellation,Safety concerns.

Demand Area Visualization

Displays simulated high and low demand zones

Driver Day Planner

Suggests best operating zones and Estimates trips and earnings

Earnings Dashboard

Daily earnings, Monthly earnings, Trend analysis.

Performance Insights

Average rating, Feedback summary, Improvement suggestions.

Technology Stack

The application is developed using lightweight front-end technologies:

Frontend:

HTML5 – Structure of the application

CSS3 – Styling and responsive design

JavaScript – Application logic and interactivity

Design Approach:

Mobile-first responsive design

Glassmorphism UI styling

Smooth animations and transitions

System Design & Architecture

The system follows a client-side architecture, where all logic is processed in the browser.

Workflow:

1. User inputs travel details
2. System simulates real-world conditions
3. Multiple travel options are generated
4. Each option is evaluated using predefined logic
5. Best option is recommended to the user

Decision Logic:

The recommendation system uses a weighted scoring approach based on:

Travel time, Cost, Traffic conditions, Convenience

Created dashboards for: Passenger and Driver

7. Feasibility Analysis

Technical Feasibility

Built using simple technologies

No dependency on external APIs

Easy to implement within given time

Operational Feasibility

Suitable for daily commuters and drivers

Easy to use interface

Economic Feasibility

Low development cost

No infrastructure required

How is the Prototype Unique?

1. Dual-Perspective Mobility System (Passenger + Driver)

Unlike conventional mobility applications that focus only on passengers, SMART C integrates both passenger and driver perspectives within a single platform.

Passengers receive optimized travel recommendations

Drivers receive tools for : demand analysis, earnings estimation, operational planning.

This creates a complete mobility ecosystem, not just a travel tool.

2. Intelligent Decision-Making Instead of Data Display

Most existing systems :

- Show routes
- Display time and cost
- SMART C goes further by:
- Analyzing multiple factors simultaneously
- Generating a recommended option
- Providing a clear explanation for the decision

This transforms the system from a data viewer → decision engine

3. Multi-Factor Optimization Model

The prototype considers multiple real-world parameters:

Time, Cost, Traffic, Weather, Convenience, Sustainability

Instead of optimizing for a single factor, the system performs balanced decision-making, making it more realistic and practical.

4. Real-Time Scenario Simulation

The application includes a dynamic simulation panel that models:

- Traffic conditions (Low / Medium / High)
- Weather (Sunny / Rainy)
- Peak hour variations

This allows users to understand how external conditions affect travel decisions, which is rarely implemented in basic mobility apps.

5. Driver-Centric Intelligence Features

- SMART C uniquely supports drivers by offering:
- Demand heatmap simulation
- Ride profitability insights
- Daily planning assistance and Performance tracking
- These features are generally not present in traditional mobility planning tools.

6. Sustainability Awareness Integration

- The prototype includes:
- CO_2 emission estimation
- Eco-friendly travel suggestions
- This aligns the system with sustainable mobility goals, adding environmental impact awareness to everyday decisions.

7. Integrated Safety and Feedback System

The platform includes: Emergency SOS functionality,Issue reporting (driver, vehicle, route),Feedback collection. This ensures user safety and continuous system improvement, making it more user-centric.

8. Lightweight Yet Scalable Architecture: Built entirely using HTML, CSS, and JavaScript, No dependency on heavy backend systems,Demonstrates that intelligent decision systems can be implemented efficiently using lightweight technologies

10. Future Enhancements: Integration with real-time APIs (maps, traffic), AI-based predictive analysis, GPS tracking,Live ride-sharing system,Include Airline and railways options.

11. Conclusion

SMART C provides a comprehensive solution to modern mobility challenges by combining: Intelligent decision-making, User-centric design, Practical real-world simulation .